QP CODE : 25900325

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP(HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective(DSE)-MG3DSECSC200-INTRODUCTION TO DATA SCIENCE

(2024ADMISSION ONWARDS)

Duration:2Hours MaximumMarks:70

Remember(K),Understand(U),Apply(A),Analyse(An),Evaluate(E),Create(C),Skill(S),Interest(I) and Appreciation(Ap)

Students should attempt at least onequestion from each course outcome to enhance their overall outcome attainability.

PartA

Very Short Answe r Questions

Answer all questions

Each question carries 2 marks

1. Illustrate the difference between nominal and ordinal data. [U] / [CO2]

   Nominal data are categories with no natural order or ranking. They are simply names or labels used to classify data. Ordinal data are categories with a meaningful order, but the differences between categories aren't necessarily equal.

2. List any two sources of secondary data. [K] / [CO2]

   1. Government publications – such as census reports, labor statistics, and economic surveys.
   2. Books, journals, and research articles – published by researchers, institutions, or organizations.
   3. Website

3. Identify the role of a business analyst in data projects. [U] / [CO1]

   A business analyst's role and responsibilities include gathering and analysing information and creating financial models using spreadsheets or any other software. They visualise data using charts and graphs and develop solution-oriented strategies by using cost-benefit analysis

4. Examine the scenarios in which you prefer Multiple Linear Regression over Simple Linear Regression and why? [A] / [CO4]

5. List the steps involved in performing EDA.(Any 2 steps) [K] / [CO3]

1. Data Collection – Gather data from relevant sources.
2. Data Cleaning – Handle missing values, remove duplicates, and correct inconsistencies.
3. Data Exploration – Examine data structure, types, and basic statistics.
4. Data Visualization – Use graphs and charts to identify patterns and relationships.
5. Feature Engineering – Create or modify variables to improve analysis.
6. Correlation Analysis – Study relationships between different variables.
7. Summarize Findings – Draw key insights and prepare the data for further modeling or analysis.

[2x5 = 10]

PartB

ShortAnswerQuestions

Answer any 6 questions

Each question carries 5 marks

6. Calculate mean and standard deviation for the data set: [200,300,400,500,600]. [A] / [CO2]

Step 1: Calculate Mean

$$\text{Mean} = \frac{\text{Sum of all values}}{\text{Number of values}}$$

$$\text{Mean} = \frac{200 + 300 + 400 + 500 + 600}{5} = \frac{2000}{5} = 400$$

✅ **Mean = 400**

Step 2: Calculate Standard Deviation

We'll find the **population standard deviation** (since no sample is mentioned).

$$\sigma = \sqrt{\frac{\sum (x_i - \bar{x})^2}{N}}$$

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
200	-200	40000
300	-100	10000
400	0	0
500	100	10000
600	200	40000

$$\text{Sum of squares} = 40000 + 10000 + 0 + 10000 + 40000 = 100000$$

$$\sigma = \sqrt{\frac{100000}{5}} = \sqrt{20000} = 141.421$$

✅ **Population Standard Deviation = 141.42 (approx)**

If you need the **sample standard deviation**, use $N - 1 = 4$:

$$s = \sqrt{\frac{100000}{4}} = \sqrt{25000} = 158.114$$

✅ **Sample Standard Deviation = 158.11 (approx)**

7. List two advantages of using cross-validation. [K] / [CO4]

Improved model reliability – It provides a more accurate estimate of model performance by testing it on multiple data splits.

Reduces overfitting – Ensures the model generalizes well to unseen data by validating on different subsets.

8. How are different types of data handled inunivariatenon-graphical EDA? [An] / [CO3]

1. Numerical Data (Quantitative)

- Continuous or discrete variables like age, income, or test scores.
- Techniques used:
  - Measures of central tendency: Mean, median, mode
  - Measures of dispersion: Range, variance, standard deviation, quartiles
  - Percentiles: To understand distribution and spread

Example:
For data [10, 20, 30, 40, 50] → mean = 30, standard deviation = 15.81

9. Illustrate the importance of preprocessing the data.(Any 5 points) [K] / [CO2]

- Ensures data quality by removing errors, inconsistencies, and duplicates.
- Handles missing values to prevent incomplete or biased analysis.
- Reduces noise and improves the accuracy of results.
- Converts categorical data into numerical form for algorithm compatibility.
- Normalizes or scales features so that no variable dominates due to unit differences.
- Helps in detecting and correcting outliers that may distort model predictions.
- Reduces dimensionality by removing irrelevant or redundant features.
- Ensures data consistency when merging data from multiple sources.
- Improves the performance and accuracy of machine learning models.
- Makes the data ready for analysis and reliable for decision-making.

10. Compare the significance of precision vs Recall in spam filtering.(Any 5 points) [U] / [CO4]

Precision vs Recall in Spam Filtering

- Precision refers to how many emails marked as *spam* are actually *spam*.
- Recall refers to how many of the *actual spam emails* are correctly identified as *spam*.
- High Precision means fewer legitimate (non-spam) emails are wrongly classified as spam.
- High Recall means most spam emails are successfully detected and filtered out.
- Precision focuses on reducing false positives (important emails wrongly sent to spam).
- Recall focuses on reducing false negatives (spam emails missed and left in the inbox).
- In spam filtering, high precision ensures user trust by not losing important emails.
- High recall ensures that the inbox stays clean and free from spam.
- There is often a trade-off — increasing recall can lower precision and vice versa.
- The best spam filter aims to balance both precision and recall for optimal performance.

11 . Discuss the importance of detecting outliers during EDA.(Any 5 points) [U] / [CO3]

.

- Outliers can distort statistical measures like mean and standard deviation.
- They may skew data distributions, affecting the accuracy of visualizations and summaries.
- Detecting outliers helps to identify data entry errors or measurement mistakes.
- Outliers can reveal important, rare events or unexpected patterns in the data.
- They may influence model training, leading to poor predictions or overfitting.
- Identifying them allows for better decision-making on whether to keep, transform, or remove them.
- Outlier detection improves the reliability and robustness of statistical and machine learning models.
- Helps in understanding data variability and assessing the quality of collected data.
- Ensures that data preprocessing and feature scaling steps are more effective.
- Overall, detecting outliers is essential for accurate, trustworthy, and

interpretable analysis.

12. Compare and contrast the "Modeling" and "Evaluation" stages of the data science life cycle. Explain how the output of one influences the other. [U] / [CO1]

Aspect	Modeling	Evaluation
Purpose	To build predictive or analytical models using prepared data.	To assess the performance and effectiveness of the developed models.
Focus	Applying machine learning or statistical algorithms to learn patterns from data.	Measuring how well the model performs using specific metrics.
Key Activities	Selecting algorithms, training models, tuning hyperparameters, and validating results.	Calculating metrics like accuracy, precision, recall, F1-score, RMSE, etc.
Input	Cleaned and transformed data from the preprocessing stage.	Predictions generated by the model and actual outcomes (ground truth).
Output	A trained model that makes predictions or classifications.	Quantitative performance scores indicating model quality.
Goal	To create a model that can generalize well to unseen data.	To determine if the model meets desired performance standards.
Techniques Used	Regression, classification, clustering, decision trees, neural networks, etc.	Cross-validation, confusion matrix analysis, ROC-AUC, error analysis, etc.
Iteration	May need to be repeated with different algorithms or parameters.	Guides whether retraining or model improvement is necessary.

13. Outline the importance of continuous monitoring in model maintenance.(Any 5 points) [An] / [CO4]

- Ensures the model remains accurate and reliable over time.
- Detects data drift — when the input data distribution changes from the original training data.
- Identifies concept drift — when relationships between input features and target outputs evolve.
- Helps in early detection of performance degradation before it impacts business outcomes.
- Enables timely retraining or model updates to maintain prediction quality.
- Supports compliance and accountability by tracking model behavior and decisions.
- Ensures the model continues to meet ethical and fairness standards.
- Provides real-time feedback for improving algorithms and system performance.
- Helps in detecting anomalies or unexpected behavior in production environments.
- Overall, continuous monitoring maintains model stability, trustworthiness, and long-term effectiveness.

[5x6 = 30]

PartC

Essay Questions

Answer any 2 questions

Each question carries 15 marks

14. Explain any five commonly used model evaluation matrix in data science. For each, mention whether it is used for regression or classification, and describe what it measures. [U] / [CO4]

1. Confusion Matrix

- **Type:** Classification
- **Measures:** A table showing:
  - **True Positives (TP):** Correctly predicted positives
  - **True Negatives (TN):** Correctly predicted negatives
  - **False Positives (FP):** Negatives incorrectly predicted as positives
  - **False Negatives (FN):** Positives incorrectly predicted as negatives
- **Use:** Provides a complete overview of a classification model's performance; basis for calculating Accuracy, Precision, Recall, and F1-score.

---

2. Accuracy

- **Type:** Classification
- **Measures:** The proportion of correct predictions (both positives and negatives) out of all predictions.
- **Formula:**

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

- **Use:** Indicates overall correctness; works best for balanced datasets.

3. Mean Absolute Error (MAE)

- **Type:** Regression
- **Measures:** The average of the absolute differences between predicted and actual values.
- **Formula:**

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

- **Use:** Represents the average error in the same units as the target; less sensitive to outliers than MSE.

---

4. Mean Squared Error (MSE)

- **Type:** Regression
- **Measures:** The average of the squared differences between predicted and actual values.
- **Formula:**

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

- **Use:** Penalizes larger errors more heavily; lower MSE indicates better model performance.

[U] / [CO3]

5. Root Mean Squared Error (RMSE)

- **Type:** Regression
- **Measures:** The square root of MSE; expresses error in the same units as the target variable.
- **Formula:**

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

- **Use:** Easier to interpret than MSE; lower RMSE indicates higher predictive accuracy.

Summary Table

Metric	Type	Measures
Confusion Matrix	Classification	Counts of TP, TN, FP, FN; overall classification performance
Accuracy	Classification	Overall correctness of predictions
MAE	Regression	Average absolute difference between predicted and actual values
MSE	Regression	Average squared difference between predicted and actual values
RMSE	Regression	Square root of MSE; error in same units as target variable

15.Break down the role of specialized EDA techniques in the Data Science lifecycle.(Explain any three techniques)

Stage 1: Data Collection → Data Understanding

Specialized EDA techniques:

- Data profiling (e.g., summary statistics, missing value analysis)
- Variable type detection (categorical, continuous, datetime)

Role:
Helps you understand the scope and quality of your raw data.
Example: Using pandas describe() or info() to detect unexpected nulls or data types.

Stage 2: Data Cleaning and Preprocessing

Specialized EDA techniques:

- Outlier detection: Boxplots, z-scores, IQR method, or DBSCAN.
- Missing data visualization: Heatmaps or bar charts (e.g., missingno library).
- Imbalance checks: For classification problems, comparing class frequencies.

Role:

- Detects data quality issues early.
- Guides data cleaning strategies (e.g., imputation vs. removal).
- Prevents model bias due to data imbalance or anomalies.

Stage 3: Data Exploration and Hypothesis Generation

Specialized EDA techniques:

- Univariate analysis: Histograms, density plots.

- Bivariate/multivariate analysis: Pairplots, correlation heatmaps, scatter matrices.
- Dimensionality reduction for visualization: PCA, t-SNE, UMAP.

Role:

- Reveals relationships and dependencies among variables.
- Suggests potential predictive features.
- Helps form hypotheses about data patterns before formal modeling.

Stage 4: Feature Engineering

Specialized EDA techniques:

- Feature importance ranking (using mutual information, ANOVA, etc.)
- Interaction analysis (cross-tabulation or 3D plots).
- Correlation analysis (detecting redundancy or multicollinearity).

Role:

- Guides feature selection and transformation decisions.
- Prevents overfitting by removing redundant or irrelevant variables.
- Improves model interpretability and performance.

Stage 5: Model Building and Validation

Even after model selection begins, EDA insights continue to shape decisions:

- Helps choose the right model family (e.g., linear vs. nonlinear).
- Informs transformations (e.g., log-scaling skewed variables).
- Aids in residual analysis post-modeling to verify assumptions (normality, homoscedasticity).

Stage 6: Communication and Insight Delivery

Specialized EDA visualization techniques:

- Interactive dashboards (Plotly, Power BI, Tableau).
- Storytelling visualizations (boxplots for distribution, heatmaps for relationships).

Role:

- Translates complex findings into clear, visual insights.
- Supports data-driven storytelling for stakeholders.

16. Explain Data Science and its scope in modern computing. [U] / [CO1]

[15x2 = 30]

Data Science is an interdisciplinary field that uses statistical methods, algorithms, and computational tools to extract meaningful insights and knowledge from data — whether structured (like databases) or unstructured (like text, images, or videos).

At its core, data science combines elements of:

- Mathematics and Statistics – to model and analyze data.
- Computer Science and Programming – to process and manage large data sets.
- Domain Expertise – to interpret results within a specific business or research context.

(Explain each of them)